



Reciprocal Relationships between Bullying Victimization and Prosocial Behavior and the Mediating Role of Self-Esteem among Chinese Boys and Girls: An RI-CLPM Study

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Abstract

Previous theory and research indicate an intricate connection between bullying victimization, self-esteem, and prosocial behavior. However, longitudinal evidence remains limited regarding the within-person reciprocal association between bullying victimization and prosocial behavior, whether their association operates indirectly through self-esteem, and whether these patterns differ between boys and girls. The current three-year longitudinal study, with a one-year interval, applied the RI-CLPM to explore the reciprocal relationship between bullying victimization and prosocial behavior, as well as the mediating role of self-esteem, separately for boys and girls. Data from 2,530 Chinese adolescents (51.40% boys; age = 13.29 ± 0.51 at T1) were analyzed. Results revealed a within-person reciprocal relationship between bullying victimization and prosocial behavior among boys only. By contrast, girls showed no direct within-person reciprocal effects. Instead, the longitudinal reciprocal relationship was mediated by girls' self-esteem. Specifically, girls' self-esteem at T2 mediated both the path from bullying victimization at T1 to prosocial behavior at T3 and the reverse path from prosocial behavior at T1 to bullying victimization at T3. These findings emphasize the need for gender-specific interventions, with a focus on promoting prosocial behavior for boys and enhancing self-esteem for girls to reduce the negative consequences of bullying victimization.

Keywords Bullying victimization · Prosocial behavior · Self-esteem · Gender differences · RI-CLPM · Chinese adolescents

Introduction

Early adolescence (i.e., ages 13–15; Colarusso, 1992) is a developmental stage marked by rapid transformations in interpersonal interactions, self-concept, and social functioning. Globally, approximately 30% to 40% of early adolescents report bullying victimization (Koyanagi et al., 2019; Smith et al., 2023; Tang et al., 2020). Bullying victimization has been linked to a range of adjustment difficulties, including decreased prosocial behavior (Xu et al., 2024). Prosocial behavior is increasingly salient during early adolescence as moral reasoning and perspective-taking capacities develop (Eisenberg et al., 2006; Fabes et al., 1999). Adolescents displaying less prosocial behavior frequently encounter heightened interpersonal difficulties, thus exacerbating their susceptibility to future bullying victimization (Fu et al., 2023). This evidence suggests that the connection between bullying victimization and prosocial behavior may

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be reciprocal. Nevertheless, there is still a lack of empirical studies on this reciprocal relationship. Thus far, just a single study has examined this association using the traditional CLPM and has identified a reciprocal relationship between bullying victimization and prosocial behavior, with this effect observed exclusively among girls (Lin et al., 2024). These findings indicate that the within-person effects linking bullying victimization and prosocial behavior are not yet fully understood and underscore the need for further investigation into potential gender differences. Moreover, the self-related mechanisms underlying this association, such as self-esteem, and potential gender differences in these processes, remain unclear. To fill these gaps, the current study employs the RI-CLPM to investigate the within-level reciprocal relationship between bullying victimization and prosocial behavior during early adolescence. It also examines the mediating role of self-esteem and tests potential gender differences in the relevant pathways.

Theoretical Framework

The trauma and victimization psychological adaptation model describes a reciprocal relationship between adverse/negative life experiences, schemas, and adaptation in victims (McCann et al., 1988). Schemas are defined as mental frameworks about the self and others across five core domains: safety, trust, power, esteem, and intimacy. These schemas help individuals organize and interpret adverse/negative life experiences and may shape specific psychological responses, including behavioral reactions such as aggressive behavior (McCann et al., 1988). Adverse/negative life experiences threaten existing positive schemas related to the self and others, thereby hindering adaptive functioning. The model also suggests that psychological maladaptation may place individuals at a greater risk of future adverse/negative life experiences.

Bullying victimization is defined as repeated harm experienced within the school environment (Olweus, 1993). Conceptually, bullying victimization is further treated as an adverse/negative life experience with potentially traumatic features, which primarily involve repeated interpersonal harm; accordingly, some scholars emphasize the necessity of examining it within a trauma framework (Idsoe et al., 2021; Morgado et al., 2025). Prolonged exposure to interpersonal harm may place victims in a chronic state of distress and helplessness (Idsoe et al., 2021), with these psychological responses accumulating over time and increasing vulnerability to mental health issues (Arseneault, 2017; Arseneault et al., 2010; Moore et al., 2017) and even developmental trauma disorder (DTD) or post-traumatic stress disorder (PTSD; Andreou et al., 2021; Guzzo et al., 2014; Idsoe et al., 2021; Li et al., 2023b; Morgado et al., 2025). The model

has been used to examine psychological responses to bullying victimization among adolescents (Chu et al., 2025).

Self-esteem is defined as an individual's attitude and assessment of self-worth (Rosenberg, 1965). Being bullied tends to develop negative schemas related to oneself, for example, low self-esteem (McCann et al., 1988). Negative self-esteem schemas often lead to increased maladaptive behavioral responses, such as engaging in self-destructive behaviors and aggressive behaviors (Baumeister & Scher, 1988; Donnellan et al., 2005). However, empirical applications of the trauma and victimization psychological adaptation model have rarely been extended to adaptive behavioral outcomes, such as prosocial behavior, defined as actions intended to benefit others (Eisenberg et al., 2006). Prosocial behavior enhances individuals' social competence and moral development, contributing to broader social stability and harmony. According to the theoretical framework, bullying victimization may undermine self-esteem, which in turn may be associated with lower levels of prosocial behavior. Lower prosocial behavior may also be linked to greater exposure to bullying victimization over time, suggesting a potentially reciprocal association.

Reciprocal Relationship between Bullying Victimization and Prosocial Behavior

Previous studies have frequently linked bullying victimization to multiple adjustment issues, including social withdrawal and decreased prosocial behavior (Cheng et al., 2025; Moore et al., 2017; Pimentel et al., 2020; Sousa et al., 2024; Xu et al., 2024). More recently, defending behavior, a specific type of prosocial behavior, was discovered to be positively correlated with bullying victimization (Tian et al., 2025). Defending behavior captures bystanders' prosocial responses in bullying contexts, with a focus on individuals' willingness to protect victims (Lambe & Craig, 2020). The current study did not distinguish among roles within bullying contexts (e.g., bystanders vs. victims) and instead examined whether bullied experiences were reciprocally associated with prosocial behavior in the overall student population.

Bullying victimization can longitudinally predict prosocial behavior. Adolescents who experience bullying tend to occupy a lower social status, have a limited number of friends, and report poorer peer relationships, possibly reducing prosocial engagement (Cook et al., 2010). A recent longitudinal study found that bullying victimization predicted a decline in prosocial behavior 18 months later (Hu et al., 2025). Evidence also reported that bullying victimization is linked to reduced prosocial behavior (Sharkey et al., 2015). These findings can be understood through the trauma and victimization psychological adaptation model (McCann et

al., 1988). Victims of bullying may change their psychological response patterns. One possible consequence is impaired social functioning, manifested as a decline in prosocial behavior. Specifically, victims often experience feelings of isolation, helplessness, and a lack of control over their situation (Wójcik et al., 2022). In such cases, victims may adopt social avoidance strategies in an attempt to evade potential threats; however, this can cause them to miss opportunities to form positive social connections and engage in social interactions, thereby reducing prosocial behavior (Cook et al., 2010; Juvonen & Graham, 2014; Ma et al., 2007).

Reduced prosocial behavior may further increase vulnerability to bullying victimization. The trauma and victimization psychological adaptation model (McCann et al., 1988) explains the reciprocal relationship between adverse/negative life experiences and psychological adaptation outcomes, suggesting that psychological adaptation outcomes can shape individuals' subsequent life experiences. In this context, adolescents who are unwilling to carry out prosocial behaviors are more vulnerable to victimization. During adolescence, prosocial behavior is crucial for social integration and peer acceptance (Guo et al., 2018), while poor prosocial functioning often signals a heightened risk of social exclusion (Casper et al., 2020). Adolescents who are introverted or socially withdrawn are more likely to be bullied due to limited peer connections and reduced social support. Previous studies have found that lower prosocial behavior was associated with subsequent physical, non-physical, and relational victimization in Chinese adolescents (Fu et al., 2023). Notably, another investigation into a similar Chinese adolescent development stage failed to identify this association (Hu & Li, 2018). The disparity is largely due to differences in research design. Neither study controlled for the impact of prior bullying victimization, highlighting the need for further investigation using the RI-CLPM. Moreover, given the limited relevant research, similar null effects are rarely documented in Western cultural contexts. For example, studies among elementary school students have shown that prosociality-based interventions can effectively prevent bullying victimization (Leadbeater et al., 2016; McCarty et al., 2016).

Existing findings and theoretical work suggest that the link between bullying victimization and prosocial behavior is complex and likely involves reciprocal influences rather than a simple unidirectional effect. However, longitudinal design exploring the reciprocal influences regarding bullying victimization and prosocial behavior remains limited. Most previous studies have adopted a unidirectional perspective, analyzing either the influence of bullying victimization on prosocial behavior or the influence of prosocial behavior on bullying victimization. To date, only one longitudinal study using a CLPM among Chinese middle school

adolescents has provided evidence for a reciprocal cycle of harmful effects (Lin et al., 2024). Clarifying this reciprocal relationship is crucial for a thorough understanding of Chinese adolescents' development. Compared with the CLPM, the RI-CLPM provides more accurate causal inferences and clarifies the directionality of associations, making it particularly suitable for examining reciprocal relationships (Hamaker et al., 2015). This demonstrates that a reexamination of the reciprocal relationship between bullying victimization and prosocial behavior at the within-person level is necessary.

Self-Esteem as a Mediator

According to the trauma and victimization psychological adaptation model (McCann et al., 1988), self-esteem may mediate the path from bullying victimization to prosocial behavior. In the model, self-esteem is described as a type of self-related schema (McCann et al., 1988). The model suggests that adolescents experiencing school bullying are more likely to suffer from poor self-esteem, which may reduce adaptive behavioral responses, such as decreased prosocial behavior. Existing evidence suggests that self-compassion may mediate this pathway (Xu et al., 2024). Self-compassion represents a salient source of positive self-regard (Neff, 2011). Moreover, research has consistently highlighted how bullying victimization negatively affects self-esteem. Specifically, a long-term study of middle school adolescents in South Korea revealed that bullying victimization was a predictor of subsequent declines in self-esteem (Choi & Park, 2021). Another longitudinal study found that peer difficulties during late childhood predicted a negative self-concept during early adolescence (Zhao et al., 2023). Studies have also highlighted the connection between self-esteem and prosocial behavior. A meta-analytic review found that declines in self-esteem may further inhibit the expression of prosocial behavior (Tsaousis, 2016). Another longitudinal study discovered that self-esteem at age 14 predicted prosocial behavior at age 17 among UK adolescents (Zhao et al., 2025a). Similar longitudinal links between self-evaluation and prosocial behavior have been reported in Chinese adolescents aged approximately 15 years (Xia et al., 2025). Prosocial behavior frequently necessitates social trust and self-efficacy, both of which are weakened in individual with poor self-esteem (Caprara et al., 2012; Carattini & Roesti, 2025).

Self-esteem may also mediate the reverse pathway, from prosocial behavior to bullying victimization. Existing evidence supports the influence of prosocial behavior on self-esteem (Zuffianò et al., 2014). This pattern complies with the sociometer theory of self-esteem, which posits that self-esteem serves as an internal indicator of social acceptance

or rejection (Leary, 2005; Leary & Baumeister, 2000). When individuals exhibit low levels of prosocial behavior, they may provoke peer rejection (Quarmley et al., 2022), which can lead to impaired self-esteem (Sandstrom & Zakriski, 2004). Recent research with Chinese adolescents also supported the finding that prosocial behavior predicts subsequent self-evaluation (Xia et al., 2025). Besides, studies have underlined that low self-esteem can predict bullying victimization (Egan & Perry, 1998; Fanti & Henrich, 2014). Negative self-evaluation can predict poor peer social functioning (i.e., peer difficulties and poor friendships) among British adolescents from ages 10 to 17 (Zhao et al., 2023). In the non-Western cultural context of South Korea, researchers also reported that self-esteem negatively predicts bullying victimization (Choi & Park, 2021). When considered collectively, these results imply that the reciprocal link between bullying victimization and reduced prosocial behavior may be mediated by self-esteem.

Gender Differences

Researchers have emphasized the importance of accounting for gender when investigating adolescent psychosocial development (Busch et al., 2015). Gender socialization theory states that girls are socialized toward helping, kindness, and harmony, with an emphasis on avoiding aggression and overt conflict, whereas boys are socialized toward toughness, independence, and competitiveness (Fagot et al., 2000; Stockard, 2006). Within this socialization context, prosocial behavior is more aligned with feminine norms. Empirical studies indicate that girls generally exhibit more prosocial behavior than boys, whereas boys display more physical aggressive behavior than girls (Björkqvist, 2018; Van Reemst et al., 2016). In addition, because harmony is highly valued in Chinese society (Cai et al., 2011; Yang et al., 2019), girls may be subject to stronger expectations to maintain prosocial behavior.

According to gender socialization theory, even when exposed to bullying, girls may still be inclined to maintain prosocial behavior in line with gender role expectations. By contrast, bullied boys are more prone to exhibit externalizing responses, which may include increased aggression as a means of self-protection (Salmivalli et al., 1996). However, empirical evidence on prosocial behavior remains mixed. Some studies suggest that both boys and girls exhibit reduced prosocial behavior following bullying victimization (Rusby et al., 2019), whereas other research documented that only girls exhibit this pattern (Lin et al., 2024). Furthermore, when facing bullying, girls may be more likely to internalize distress than boys. Boys, in contrast, may use externalizing coping strategies more frequently, such as overt confrontation or retaliation. For girls, however, such

externalizing responses may conflict with gender socialization that emphasizes compliance and harmony, which may be linked to a poorer sense of self-worth. Research has found that girls are more affected by bullying victimization in terms of their self-esteem (Zhang et al., 2024). Such a state of lowered self-esteem may translate into reduced prosocial behavior, as the expression of girls' prosocial behavior is particularly driven by internal emotional states (Fabes et al., 1999; Hastings et al., 2006).

Similarly, given that girls are socialized toward kindness and harmony and generally display more helping and cooperative behaviors, they could be less inclined to bully others and more empathetic (Fagot et al., 2000; Stockard, 2006). This greater inclusiveness may allow them to tolerate reductions in an individual's prosocial behavior within peer groups, thereby making such reductions less likely to directly elicit subsequent bullying victimization among girls. Previous research indicates that bullying is often less common among girls than boys, with this gender difference demonstrating cross-cultural consistency (Smith et al., 2019). By contrast, among boys, declines in prosocial behavior may be less readily accepted within male peer groups, as pro-sociality is closely tied to peer acceptance and social competence, and male peer contexts tend to exhibit higher overall levels of bullying behavior. Together, these dynamics may render boys with lower prosocial behavior particularly vulnerable to subsequent bullying victimization. Previous research reported that boys with limited prosocial competencies, heightened emotional difficulties, and broader problems in social functioning, as well as symptoms of hyperactivity, are the most vulnerable to bullying victimization (Johnson et al., 2002). However, lower levels of prosocial behavior may be particularly detrimental to girls' self-esteem, as deviating from socially prescribed expectations of kindness and harmony may elicit self-evaluative distress and a perceived failure to meet normative standards. Consequently, girls' self-worth is compromised. Evidence suggests the influence of prosocial behavior on self-esteem was observed only among girls (Laible et al., 2004). In addition, findings regarding the link between self-esteem and subsequent victimization remain inconsistent. Some studies suggest that self-esteem more strongly predicts subsequent victimization among boys (Cava et al., 2010; Zhao et al., 2021), whereas other work indicates that low self-esteem increases the likelihood of later victimization among girls (Prinstein et al., 2001). Even with these insights, little literature has systematically examined gender differences in adolescent bullying victimization, self-esteem, and prosocial behavior within a reciprocal framework.

Current Study

Based on the existing evidence, several gaps remain in research on the relationship between bullying victimization and prosocial behavior. First, evidence is inconclusive as to whether bullying victimization and prosocial behavior are reciprocally related. Moreover, the potential mediating role of self-esteem in this relationship has yet to be explored, and how these processes differ between boys and girls remains unclear. Furthermore, most prior studies have focused on between-person effects, with limited evidence on within-person effects. The primary aim of the current study is to address these gaps by utilizing three-wave longitudinal data and applying the RI-CLPM. Several hypotheses were tested in the current study. At the between-person level, bullying victimization is related to prosocial behavior; at the within-person level, the association between bullying victimization and prosocial behavior is reciprocal (Hypothesis 1). At the between-person level, bullying victimization, self-esteem, and prosocial behavior are interrelated; at the within-person level, self-esteem serves as a longitudinal mediator in both directions, linking bullying victimization to subsequent prosocial behavior and prosocial behavior to subsequent bullying victimization (Hypothesis 2). Additionally, these associations may differ by gender (Hypothesis 3).

Methods

Participants

A total of 2,530 adolescents (51.42% boys; $M_{\text{age}}=13.29$, $SD_{\text{age}}=0.51$) from 13 middle schools in Beijing, China, were recruited for the current longitudinal study. Among the 13 middle schools included in this study, 7 were located in the urban core districts, and 6 were located outside the urban core districts. Data were collected across three waves (T1–T3) at one-year intervals. At T1, 2,473 participants completed the survey (51.48% boys; $M_{\text{age}}=13.29$, $SD_{\text{age}}=0.51$). They represented 97.75% of the total sample. Between T1 and T2, 340 participants dropped out while 53 new participants were added, and the number of participants at T2 was 2,186 (50.64% boys; $M_{\text{age}}=14.26$, $SD_{\text{age}}=0.48$). They represented 86.40% of the total sample. Between T2 and T3, 417 participants dropped out while 45 new participants were added, and the number of participants at T3 was 1,814 (50.61% boys; $M_{\text{age}}=15.25$, $SD_{\text{age}}=0.44$). They represented 71.70% of the total sample. Transferring to a different school or missing the assessment day were the main causes of attrition. To enhance data utilization, the subsequent analyses included the total sample, comprising data from 2,530 adolescents.

Measures

Bullying victimization

Bullying victimization was assessed using the School Bullying Victimization Scale (Dong & Lin, 2011). This measure was developed by Dong and Lin (2011) as a shorter version of the Chinese Olweus Bully/Victim Questionnaire (Zhang et al., 1999). This measure consists of 7 items that assess the frequency of bullying victimization experiences. A sample item is “Being teased or picked on”. Participants responded on a 5-point scale ranging from 1 (never happened) to 5 (five or more times). The mean score was computed. This measure has been widely verified (Guo et al., 2022; Ren et al., 2023; Xiong et al., 2023). The scale in the current study demonstrated good reliability across the three waves (Cronbach’s $\alpha=0.874$ at T1, 0.867 at T2, and 0.878 at T3). Additionally, the CFA results at three waves are provided in the supplementary materials (Table S1) and indicate good validity.

Self-Esteem

Self-esteem was measured using the Rosenberg Self-Esteem Scale (Rosenberg, 1965). The Rosenberg Self-Esteem Scale consists of 10 items, including 4 reverse-coded items. A sample item is “I take a positive attitude toward myself”. Participants rated each item on a 4-point scale ranging from 1 (strongly disagree) to 4 (strongly agree). After reverse coding, the mean score of all items was computed. This measure has been widely verified (Yang et al., 2024). The scale in the current study demonstrated good reliability across the three waves (Cronbach’s $\alpha=0.849$ at T1, 0.868 at T2, and 0.884 at T3).

Prosocial Behavior

Prosocial behavior was assessed using the Prosocial Behavior Scale (Dong & Lin, 2011). It consists of 12 items that assess prosocial behavior across four domains: rule-related, trait-related, relationship-related, and altruism-related prosocial behavior. A sample item is “I offer to help others when they are in need”. Participants rated each item on a 4-point scale ranging from 1 (never) to 4 (always). The mean score of all items was computed to yield an overall index of general prosocial behavior. This measure has been widely verified (Wang et al., 2023; Zhou et al., 2023). The scale showed excellent reliability across the three waves (Cronbach’s $\alpha=0.924$ at T1, 0.927 at T2, and 0.933 at T3). Additionally, the CFA results at three waves are provided in the supplementary materials (Table S2) and indicate good validity.

Demographic and Control Variable

Adolescents reported their school, gender, age, parents' educational level, occupational level, and family possessions. Controlling for family socioeconomic status (SES) and age to account for potential confounding (Yang et al., 2025; Zheng et al., 2025). SES was computed based on parents' educational level, occupational level, and family possessions (Lee et al., 2019). A principal component analysis (PCA) yielded a single factor that accounted for 62.93% of the variance. SES scores were calculated using the following formula: $SES = (0.879 \times \text{parents' average educational level} + 0.826 \times \text{parents' occupation} + 0.657 \times \text{family ownership}) / 0.6293$. According to developmental contextualism (Lerner, 1998), development results from the combined influence of individual characteristics and environmental experiences. Adolescents spend a large amount of their time in school every day, making the school context crucial to their development. Bullying and victimization primarily occur in schools, where the school norms, peer climate, and teacher practices can strongly influence these experiences (Låftman et al., 2017). Similarly, the development of self-esteem and prosocial behavior is closely linked to the school environment (Li et al., 2021; Li et al., 2023a; Luo et al., 2023). As a result, controlling for school impacts is essential. Given the relatively small number of schools, the multilevel RI-CLPM could not converge. Therefore, school was coded as a dummy variable and included as a covariate together with age and SES by regressing the random intercepts on these variables.

Procedure

All data were drawn from a Longitudinal Survey of Adolescents' Sociability Development in China, which was approved by the Research Ethics Committee of the Collaborative Innovation Center of Assessment toward Basic Education Quality at Beijing Normal University. The parents of the adolescents provided written informed permission. To ensure standardized procedures, the investigation was conducted by psychology graduate students with training. Students were informed prior to the start of the test that they might withdraw at any moment and that their answers would only be utilized for scientific study. Three measurements were conducted during school class time. Over the course of the three data collection rounds, each participant received a unique code, which was used to link their data over the three years.

Statistical Analyses

Preliminary analyses, including missing value analysis and imputation, descriptive statistics, correlation analysis, and intraclass consistency coefficients (ICCs), were conducted using SPSS 26.0. Measurement invariance analysis, the RI-CLPM analysis, and mediation analysis were conducted using Mplus 8.3. Using multiple indicators in longitudinal models can increase model complexity and complicate estimation (Little et al., 2022). To ensure model convergence, mean scores rather than multiple indicators were used in the RI-CLPM analyses, as well as in the RI-CLPM models with the mediator.

We first conducted analyses of missing data and applied appropriate handling procedures. To understand the pattern of missing data, Little's MCAR test was conducted, and the results showed a standardized $\chi^2/df = 1.303 (<3)$, indicating that the pattern of missing data was not substantially different from the random pattern (Bollen, 1989). To further evaluate potential attrition bias, we examined whether there were significant differences between participants with and without missing data at T2 or T3 on main variables at T1. Results showed no significant differences for T1 bullying victimization ($M_{\text{dropout}} = 1.84$, $SD_{\text{dropout}} = 0.95$, $M_{\text{stay}} = 1.77$, $SD_{\text{stay}} = 0.92$, $p = 0.175$) and T1 self-esteem ($M_{\text{dropout}} = 3.12$, $SD_{\text{dropout}} = 0.58$, $M_{\text{stay}} = 3.13$, $SD_{\text{stay}} = 0.56$, $p = 0.840$). However, a significant difference was found for T1 prosocial behavior ($M_{\text{dropout}} = 3.16$, $SD_{\text{dropout}} = 0.66$, $M_{\text{stay}} = 3.29$, $SD_{\text{stay}} = 0.61$, $p = 0.002$), indicating that dropout rates were higher among those who exhibited less prosocial behavior. The effect size was 0.62, which falls in the medium range (Cohen, 1988). To maximize data retention (Wang et al., 2025; Xia et al., 2025), the EM algorithm was employed to handle missing values (Schafer & Graham, 2002). Therefore, all 2,530 participants were included in the analysis.

Second, descriptive statistics and bivariate correlations for all study variables were computed. Then, ICCs were calculated for all study variables to examine the proportion of variance at the within-person (i.e., intraindividual fluctuations over time) and between-person (i.e., interindividual differences) levels. If at least 10% of the variance was attributable to the within-person level, RI-CLPM analyses were conducted (Hamaker et al., 2015). Specifically, 40.7%–57.1% of the variance was attributable to between-person effects, while 42.9%–59.3% was attributable to within-person effects, suggesting that it is possible and necessary to separate between and within effects.

Third, longitudinal measurement invariance and multi-group measurement invariance were assessed to establish structure, weak, and strong equivalence across time waves and gender groups. Model comparisons were conducted to examine whether imposing constraints significantly reduced

model fit relative to the unconstrained model. Considering the sensitivity of chi-square values in large samples (Bentler & Bonett, 1980), a decrease in $CFI \leq 0.01$ and an increase in $RMSEA \leq 0.015$ were taken to indicate that the constrained model did not meaningfully reduce model fit, supporting measurement equivalence (Chen, 2007). The strong equivalence was established across three waves and gender. Detailed results of measurement invariance are provided in the supplementary materials (Table S3 and Table S4).

Then, multigroup RI-CLPM was performed to examine within-person associations over time, their directionality, and potential gender differences. Specifically, a baseline model was estimated in which all parameters were freely estimated for boys and girls separately. Subsequently, a constrained model was specified with gender equality constraints on both the between-person effects (i.e., correlations and variances of the random intercepts) and the within-person effects. The model used to address hypothesis 1 included bullying victimization and prosocial behavior experienced from T1 to T3. The model used to address hypothesis 2 was further extended to include self-esteem from T1 to T3. Gender differences were examined to address hypothesis 3 by comparing nested models using a chi-square difference test. Effect size in the RI-CLPM was evaluated using the standardized path coefficient according to the criteria proposed by Orth et al. (2024), with the cutoffs for small, medium, and large effects of 0.03, 0.07, and 0.12, respectively.

Finally, to evaluate whether self-esteem accounted for both indirect pathways linking bullying victimization and prosocial behavior, bootstrap (5,000 samples) was used to estimate 95% CI for within-person mediation effects. The standardized indirect effect was used to determine the mediation effect's effect size. Following the guidelines proposed by Preacher and Kelley (2011), 0.01 corresponds to a small effect, 0.09 to a medium effect, and 0.25 to a large effect. For an example of a study applying the same standard, see Jeong et al. (2020). Structural equation model fit was evaluated using four indices: $\chi^2/df \leq 5$, $CFI \geq 0.90$, $TLI \geq 0.90$, and $RMSEA \leq 0.08$, as suggested by Wen et al. (2004).

To understand the potential bias of the missing data handling approach (EM algorithm), we conducted sensitivity analyses using listwise deletion following the approach of previous studies (Li et al., 2025; Liang & Chen, 2024; Zhou et al., 2025). Specifically, we excluded all participants who dropped out at any of the three time points and re-examined the hypothesized models. The reciprocal relation between bullying victimization and prosocial behavior for boys, although it was not significant at the 95% level, still showed a medium effect size. In addition, for girls, the reciprocal mediating effects of self-esteem remained consistent, showing medium to large effect sizes. The results obtained from the listwise-deleted sample were largely consistent with

those based on the full sample using the EM algorithm. Detailed results of the sensitivity analysis are provided in the supplementary materials.

Results

Descriptive Statistics and Correlations

Means, standard deviations, and correlations among bullying victimization (BV), self-esteem (SE), and prosocial behavior (PB) across three waves are presented in Table 1. Significant positive correlations in BV were observed across the three survey waves, and similar positive correlations were observed for SE and PB. Over the three waves, BV was significantly negatively associated with both SE and PB, whereas SE was significantly positively associated with PB. Gender was significantly negatively correlated with BV at T1, T2, and T3, indicating that boys reported higher levels of BV than girls. Gender was significantly positively correlated with SE at T1, suggesting that girls reported higher SE than boys. Gender was also significantly positively correlated with PB at all three waves, indicating that girls demonstrated higher levels of PB than boys. Age was significantly positively correlated with BV at all three waves, and significantly negatively correlated with SE at all three waves, as well as with PB at T2 and T3. SES was significantly negatively correlated with BV at all three waves, and was significantly positively correlated with SE and PB at all three waves. These correlations support the appropriateness of employing the RI-CLPM, as well as the necessity of conducting gender-specific analyses.

Testing Reciprocal Associations between Bullying Victimization and Prosocial Behavior among Boys and Girls

Results showed that the model significantly differed between boys and girls ($\chi^2_{\text{baseline model}} = 304.666$, $df = 114$, $\chi^2/df = 2.673$, $CFI = 0.949$, $TLI = 0.911$, $RMSEA = 0.036$; $\chi^2_{\text{constraint model}} = 398.741$, $df = 125$, $\chi^2/df = 3.190$, $CFI = 0.927$, $TLI = 0.884$, $RMSEA = 0.042$; $\Delta\chi^2 = 94.075$, $\Delta df = 11$, $p < 0.001$). Subsequently, we performed separate analyses for boys and girls.

Model 1_{boy} fitted the data well ($\chi^2 = 174.366$, $df = 57$, $\chi^2/df = 3.059$, $CFI = 0.927$, $TLI = 0.873$, $RMSEA = 0.040$). The RI_{BV} and the RI_{PB} was significantly and negatively correlated ($\beta = -0.379$, 95%CI = [-0.517, -0.240]). At the within-person level, significant autoregressive paths were observed for PB: the path from T1 to T2 was significant ($\beta = 0.156$, 95%CI = [0.070, 0.242]), and the path from T2 to T3 was significant ($\beta = 0.200$, 95%CI = [0.096, 0.303]). All significant

Table 1 Descriptive statistics and correlations among all variables ($N_{\text{Total}}=2530$; $N_{\text{Boys}}=1301$; $N_{\text{Girls}}=1229$)

Variable	1	2	3	4	5	6	7	8	9	10	11	12
1. Gender	—											
2. Age	-0.116***	—										
3. SES	0.040*	-0.213***	—									
4. T1 BV	-0.220***	0.072***	-0.113***	—								
5. T2 BV	-0.197***	0.097***	-0.079***	0.414***	—							
6. T3 BV	-0.192***	0.071***	-0.076***	0.391***	0.489***	—						
7. T1 SE	0.048*	-0.072***	0.131***	-0.267***	-0.194***	-0.166***	—					
8. T2 SE	0.03	-0.108***	0.155***	-0.272***	-0.322***	-0.245***	0.581***	—				
9. T3 SE	-0.018	-0.091***	0.133***	-0.219***	-0.211***	-0.237***	0.491***	0.648***	—			
10. T1 PB	0.100***	-0.039*	0.158***	-0.215***	-0.202***	-0.165***	0.389***	0.325***	0.237***	—		
11. T2 PB	0.067***	-0.094***	0.134***	-0.196***	-0.248***	-0.205***	0.352***	0.404***	0.303***	0.475***	—	
12. T3 PB	0.057**	-0.061**	0.075***	-0.170***	-0.150***	-0.175***	0.270***	0.328***	0.379***	0.454***	0.535***	—
$M \pm SD_{\text{Total}}$	—	13.29 $\pm$ 0.51	0.01 $\pm$ 2.95	1.79 $\pm$ 0.92	1.54 $\pm$ 0.72	1.40 $\pm$ 0.60	3.12 $\pm$ 0.55	3.12 $\pm$ 0.55	3.21 $\pm$ 0.50	3.26 $\pm$ 0.62	3.24 $\pm$ 0.61	3.30 $\pm$ 0.55
$M \pm SD_{\text{Boy}}$	—	13.35 $\pm$ 0.55	-0.10 $\pm$ 2.89	1.99 $\pm$ 1.03	1.68 $\pm$ 0.81	1.51 $\pm$ 0.72	3.10 $\pm$ 0.57	3.11 $\pm$ 0.56	3.22 $\pm$ 0.49	3.20 $\pm$ 0.66	3.20 $\pm$ 0.64	3.27 $\pm$ 0.56
$M \pm SD_{\text{Girl}}$	—	13.23 $\pm$ 0.47	0.13 $\pm$ 3.00	1.58 $\pm$ 0.74	1.40 $\pm$ 0.56	1.28 $\pm$ 0.43	3.15 $\pm$ 0.54	3.14 $\pm$ 0.53	3.20 $\pm$ 0.50	3.32 $\pm$ 0.56	3.28 $\pm$ 0.57	3.33 $\pm$ 0.53

Note. Gender was dummy coded (1: boy; 2: girl). BV=bullying victimization; SE=self-esteem; PB=prosocial behavior. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

autoregressive paths demonstrated large effect sizes. Regarding cross-lagged paths, higher T1 BV exerted a significant effect on lower T2 PB ($\beta = -0.081$, 95%CI=[-0.155, -0.008]), while lower T2 PB exerted a significant effect on higher T3 BV ($\beta = -0.133$, 95%CI=[-0.236, -0.029]). Every significant cross-lagged path showed effect sizes ranging from medium to large, and no other cross-lagged paths reached significance. Please see detailed in Fig. 1.

Model 1_{girl} fitted the data well ($\chi^2 = 130.300$, $df = 57$, $\chi^2/df = 2.286$, CFI=0.966, TLI=0.940, RMSEA=0.032). The RI_{BV} and the RI_{PB} was significantly and negatively correlated ($\beta = -0.205$, 95%CI=[-0.296, -0.114]). At the within-person level, significant autoregressive paths were observed: for BV, the path from T1 to T2 was significant ($\beta = 0.191$, 95%CI=[0.119, 0.263]); for PB, the path from T2 to T3 was significant ($\beta = 0.111$, 95%CI=[0.002, 0.221]). All significant autoregressive paths showed effect sizes ranging from medium to large, and no cross-lagged paths between BV and PB reached significance. Please see detailed in Fig. 2.

Testing the Mediation Role of Self-Esteem among Boys and Girls

Results showed that the model significantly differed between boys and girls ($\chi^2_{\text{baseline model}} = 400.346$, $df = 174$, $\chi^2/df = 2.301$, CFI=0.970, TLI=0.944, RMSEA=0.032; $\chi^2_{\text{constraint model}} = 545.582$, $df = 198$, $\chi^2/df = 2.755$, CFI=0.954, TLI=0.924, RMSEA=0.037; $\Delta\chi^2 = 145.236$, $\Delta df = 24$, $p < 0.001$). Subsequently, we separately performed the analysis for boys and girls.

Model 2_{boy} fitted the data well ($\chi^2 = 234.411$, $df = 87$, $\chi^2/df = 2.694$, CFI=0.958, TLI=0.921, RMSEA=0.036). The RI_{BV} was significantly and negatively associated with the RI_{SE} ($\beta = -0.412$, 95%CI=[-0.583, -0.250]) and the RI_{PB} ($\beta = -0.376$, 95%CI=[-0.587, -0.197]). In contrast, the RI_{SE} was significantly and positively related to the RI_{PB} ($\beta = 0.477$, 95%CI=[0.290, 0.609]). At the within-person level, significant autoregressive paths were observed: for SE, the path from T1 to T2 was significant ($\beta = 0.288$, 95%CI=[0.188, 0.371]) and the path from T2 to T3 ($\beta = 0.384$, 95%CI=[0.256, 0.501]) was significant; for PB, the path from T1 to T2 was significant ($\beta = 0.105$, 95%CI=[0.007, 0.206]) and the path from T2 to T3 ($\beta = 0.193$, 95%CI=[0.078, 0.312]) was significant. All significant autoregressive paths showed effect sizes ranging from medium to large. Regarding cross-lagged paths, higher BV at T1 exerted a significant effect on lower SE at T2 ($\beta = -0.110$, 95%CI=[-0.186, -0.034]); lower SE at T1 exerted a significant effect on lower PB at T2 ($\beta = 0.197$, 95%CI=[0.102, 0.296]); and lower PB at T1 exerted a significant effect on lower SE at T2 ($\beta = 0.106$, 95%CI=[0.027, 0.192]). All significant cross-lagged paths

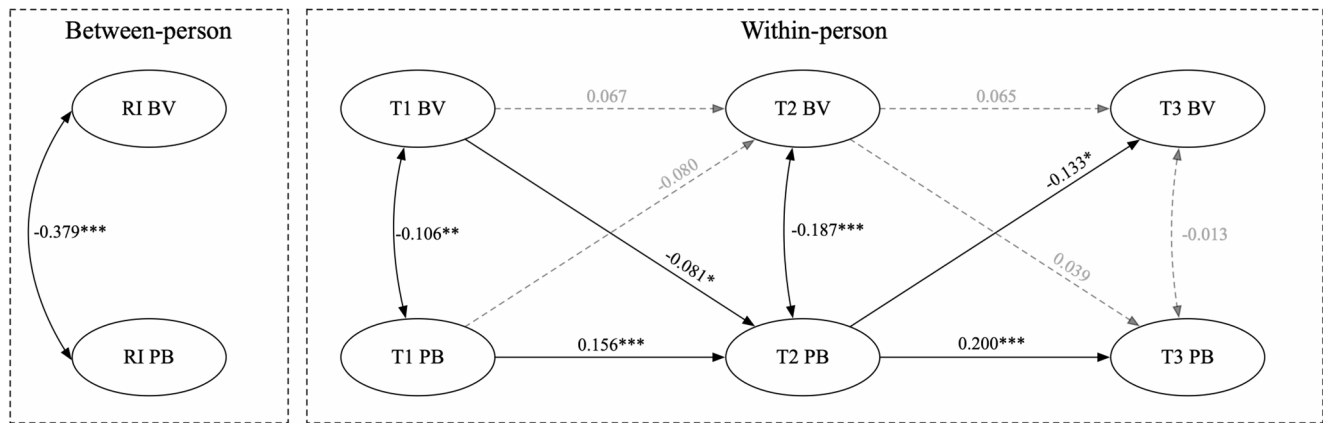


Fig. 1 Model 1_{boy}: Standardized estimates of the RI-CLPM for boy. Note. SES, age and school were also included in the model as control variables but were not presented for clarity. Dotted paths indicated

that the coefficients were not significant. BV=bullying victimization; SE=self-esteem; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

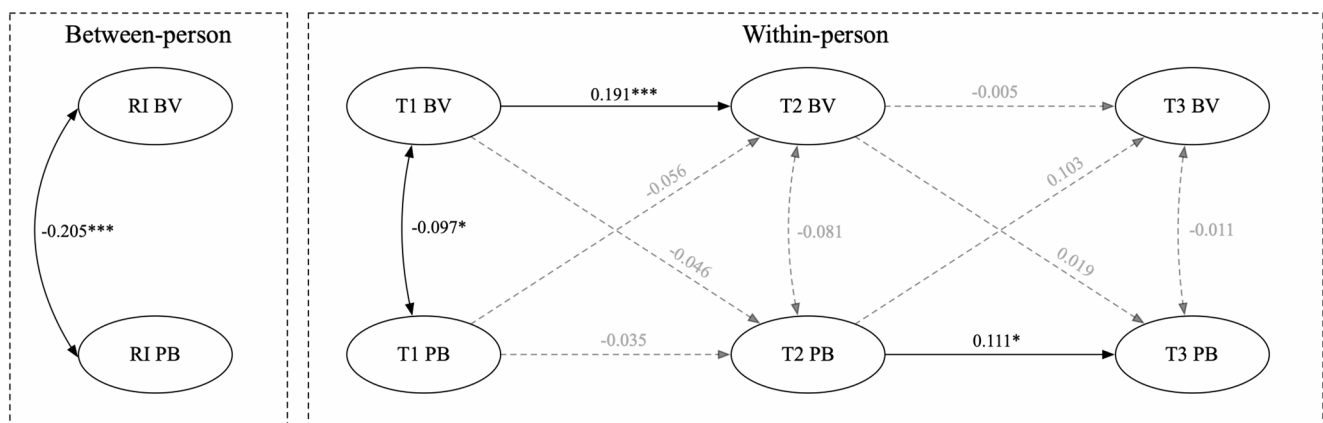


Fig. 2 Model 1_{girl}: Standardized estimates of the RI-CLPM for girl. Note. Refer to Fig. 1

showed effect sizes ranging from medium to large, and no other cross-lagged paths reached significance. Please see detailed in Fig. 3.

Further mediation analyses indicated that T2 SE did not mediate the path from T1 BV to T3 PB (indirect effect_{T1 BV→T2 SE→T3 PB}=−0.009, 95%CI=[−0.029, 0.001]), nor T1 PB to T3 BV (indirect effect_{T1 PB→T2 SE→T3 BV}=−0.009, 95%CI=[−0.031, 0.002]). Please see detailed in Table 2.

Model 2_{girl} fitted the data well ($\chi^2=165.935$, $df=87$, $\chi^2/df=1.907$, CFI=0.980, TLI=0.963, RMSEA=0.027). The RI_{BV} was significantly and negatively associated with the RI_{SE} ($\beta=-0.263$, 95%CI=[−0.419, −0.089]) and the RI_{PB} ($\beta=-0.178$, 95%CI=[−0.316, −0.048]). In contrast, the RI_{SE} was significantly and positively related to the RI_{PB} ($\beta=0.415$, 95%CI=[0.281, 0.518]). At the within-person level, significant autoregressive paths were observed: for BV, the path from T1 to T2 was significant ($\beta=0.219$, 95%CI=[0.122, 0.316]); for SE, the path from T1 to T2 was significant ($\beta=0.238$, 95%CI=[0.113, 0.356]) and

the path from T2 to T3 was also significant ($\beta=0.396$, 95%CI=[0.259, 0.501]). All significant autoregressive paths demonstrated large effect sizes. Regarding cross-lagged paths, from T1 to T2, higher BV at T1 exerted a significant effect on lower SE at T2 ($\beta=-0.172$, 95%CI=[−0.253, −0.093]); lower SE at T1 exerted a significant effect on lower PB at T2 ($\beta=0.276$, 95%CI=[0.171, 0.385]); and in turn, lower PB at T1 exerted a significant effect on lower SE at T2 ($\beta=0.130$, 95%CI=[0.033, 0.221]). From T2 to T3, higher BV at T2 exerted a significant effect on lower SE at T3 ($\beta=-0.134$, 95%CI=[−0.242, −0.016]); conversely, lower SE at T2 also exerted a significant effect on higher BV at T3 ($\beta=-0.193$, 95%CI=[−0.384, −0.032]). In addition, lower SE at T2 exerted a significant effect on lower PB at T3 ($\beta=0.292$, 95%CI=[0.173, 0.417]), and lower PB at T2 also exerted a significant effect on lower SE at T3 ($\beta=0.125$, 95%CI=[0.041, 0.212]). All significant cross-lagged paths showed large effect sizes, and no other cross-lagged paths reached significance. Please see detailed in Fig. 4.

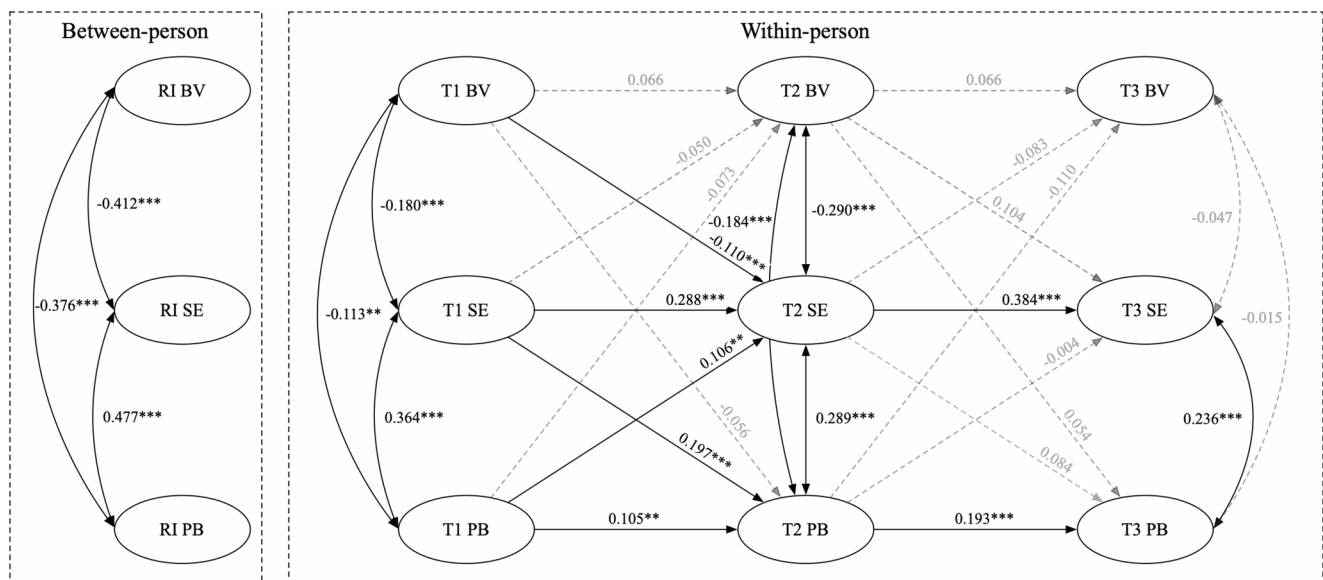


Fig. 3 Model 2_{boy}: Standardized estimates of the RI-CLPM with mediator for boy. *Note.* Refer to Fig. 1

Table 2 Standardized mediation effects

Indirect Path	Boys		Girls	
	Indirect effect	95%CI	Indirect effect	95%CI
T1 BV→T2 SE→T3 PB	-0.009	[-0.029, 0.001]	-0.050	[-0.087, -0.025]
T1 PB→T2 SE→T3 BV	-0.009	[-0.031, 0.002]	-0.025	[-0.069, -0.003]

Note. BV=bullying victimization; SE=self-esteem; PB=prosocial behavior

Further mediation analysis indicated that T2 SE simultaneously mediated the path from T1 BV to T3 PB (indirect effect_{T1 BV→T2 SE→T3 PB}= -0.050 , 95%CI=[-0.087 , -0.025]), as

well as T1 PB to T3 BV (indirect effect_{T1 PB→T2 SE→T3 BV}= -0.025 , 95%CI=[-0.069 , -0.003]). Both reciprocal longitudinal mediation effects demonstrated large effect sizes. Please see detailed in Table 2.

Discussion

Prior research provides limited insight into the reciprocal association between bullying victimization and prosocial behavior, the mediating role of self-esteem in both directions, and potential gender differences. Moreover, most studies rely on unidirectional or between-person models,

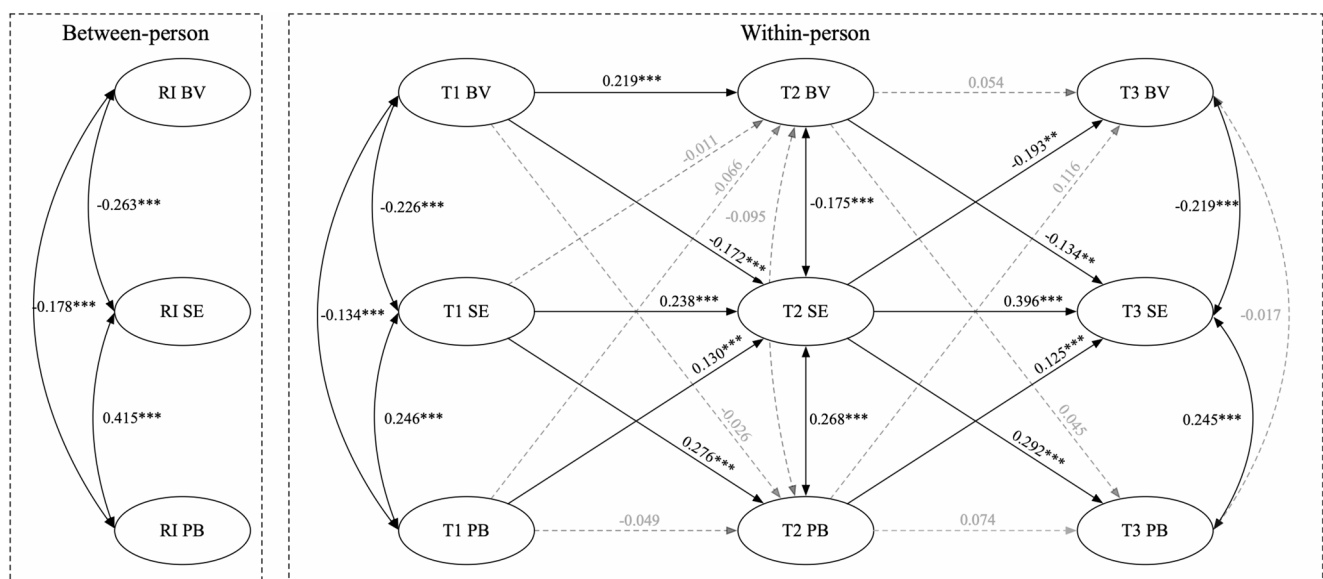


Fig. 4 Model 2_{girl}: Standardized estimates of the RI-CLPM with mediator for girl. *Note.* Refer to Fig. 1

which leave within-person effects unclear. This study aimed to explore the reciprocal association between bullying victimization and prosocial behavior, the mediating role of self-esteem, and potential gender differences using a multi-group RI-CLPM among Chinese boys and girls. This study demonstrated the reciprocal link between bullying victimization and prosocial behavior only in boys; however, only girls observed the reciprocal indirect pathway through self-esteem.

Reciprocal Associations between Bullying Victimization and Prosocial Behavior among Boys and Girls

The hypothesized reciprocal effect between bullying victimization and prosocial behavior was observed only in boys. Specifically, bullying victimization at T1 reduced boys' willingness to engage in prosocial behavior at T2, and a lower level of prosocial behavior at T2 increased the risk of bullying victimization at T3. These findings extended the trauma and victimization psychological model (McCann et al., 1988), highlighting a vicious cycle between bullying victimization as an adverse/negative life experience and prosocial behavior in boys. The findings suggest that when experiencing school bullying, boys are more likely to reduce adaptive social behavior and to decrease prosocial behavior, and those who engaged less in helping and cooperative behaviors faced an elevated risk of being victimized by peers. This reciprocal relationship was not supported in girls. In contrast to the findings of Lin et al. (2024), but are consistent with gender socialization theory, which suggests that girls may face stronger normative expectations to maintain prosocial behavior (Fagot et al., 2000; Stockard, 2006).

Even when exposed to bullying, girls may still be inclined to maintain prosocial behavior because femininity norms emphasize empathy and harmony (Fagot et al., 2000; Ringrose & Renold, 2010; Stockard, 2006). Such norms may also make girls more likely to respond to victimization through empathy, which aligns with the altruism born of the suffering perspective. Specifically, individuals exposed to traumatic events may develop specific emotional responses that enhance their motivation to help others (Staub & Vollhardt, 2008; Vollhardt, 2009). Existing research indicates a positive association between trauma-related adversity and prosocial behavior (Frazier et al., 2013). Bullying victimization, considered a form of repeated interpersonal trauma (Idsoe et al., 2021), may similarly foster prosocial behavior among victims, as helping others can provide meaning to their suffering (Jolliffe & Farrington, 2011; Kokkinos & Kipritsi, 2018), especially among girls. Such processes may weaken the adverse effects of bullying victimization on prosocial behavior among girls.

Conversely, prosocial behavior also predicted subsequent bullying victimization in boys, aligning with Johnson et al. (2002), who suggested that boys with limited prosocial skills are particularly vulnerable to bullying. This vulnerability may stem from boys exhibiting low prosocial behavior compared with girls (Van der Graaff et al., 2018). Boys with low prosocial behavior may appear less integrated within peer networks and may provoke conflicts with their peers, thereby increasing their risk of being bullied. Compared to boys, girls are socialized toward relational and harmonious interactions (Fagot et al., 2000; Stockard, 2006), and they generally exhibit more prosocial behavior (Van der Graaff et al., 2018). Even if a girl's prosocial behavior decreases, it may not necessarily be interpreted by other girls as a lack of friendliness or as being difficult to get along with. Consequently, among girls, lower prosocial behavior does not appear to increase the risk of being bullied.

The Mediation Role of Self-Esteem among Boys and Girls

Across both boys and girls, bullying victimization at T1 predicted lower self-esteem at T2; lower self-esteem at T1 predicted reduced prosocial behavior at T2; and lower prosocial behavior at T1 predicted lower self-esteem at T2. In addition, only in girls, several later-wave associations emerged: bullying victimization at T2 predicted lower self-esteem at T3; lower self-esteem at T2 predicted reduced prosocial behavior at T3 and a higher risk of bullying victimization at T3; and lower prosocial behavior at T2 predicted lower self-esteem at T3. The longitudinal mediation pathway of self-esteem aligns with the trauma and victimization psychological model (McCann et al., 1988), which suggests that the adverse/negative life experience (i.e., bullying victimization) triggers negative schemas related to oneself (i.e., lower self-esteem). Negative self-esteem leads to psychological adaptation problems (i.e., lower prosocial behavior). These findings converge with existing evidence (Fu et al., 2017; Mawlod et al., 2024; Tsaousis, 2016).

Further indirect effect through self-esteem was significant among girls but not among boys. Girls' self-esteem is more closely connected to interpersonal relationships, social evaluation, and emotional connectedness (Harter, 1999). When girls experience bullying, they are more susceptible to internalizing the experience, engaging in negative self-appraisal, and attributing the cause of victimization to themselves (Rose & Rudolph, 2006; Sesar et al., 2016). In contrast, when confronted with bullying-related distress, boys may be more prone to employ externalizing coping strategies, such as aggression (Sesar et al., 2016). Related research has similarly documented that bullying victimization exerts a greater impact on self-esteem among girls

(Zhang et al., 2024). Moreover, internal emotional states like empathy and self-worth may have a greater impact on girls' prosocial behavior, whereas boys' prosocial behavior may be more affected by external factors, including social norms and reinforcement (Fabes et al., 1999; Hastings et al., 2006). Considering these aspects together, girls' self-esteem plays a mediating role between bullying victimization and prosocial behavior.

With respect to both the prospective association from prosocial behavior to self-esteem and the association from self-esteem to bullying victimization, the present study yielded findings that diverged from prior research. Laible et al. (2004) documented that lower prosocial behavior primarily predicts reduced self-esteem among girls, whereas our results showed the presence of this prediction across genders. Zhao et al. (2021) reported that low self-esteem was related to greater susceptibility to being bullied only among boys. However, our results showed this effect was observed exclusively among girls; consistent with this pattern, prosocial behavior was related to bullying victimization via self-esteem among girls, whereas no such mediating effect was observed for boys. Early studies indicated that girls who experience difficulties in psychological adjustment, such as depression, loneliness, or low self-esteem, may be more vulnerable to subsequent victimization (Prinstein et al., 2001). This pattern may be explained by gender differences in rejection sensitivity, with girls expressing higher rejection sensitivity than boys (Erozkan, 2009; Maiolatesi et al., 2022). Prior studies have demonstrated that low self-esteem has been proven to connect with more rejection sensitivity (Zhou et al., 2020), which has been further reported to connect with bullying victimization (Gao et al., 2021). This means that, among girls, low self-esteem may foster heightened rejection sensitivity, which in turn may increase vulnerability to subsequent bullying victimization. By contrast, this pathway appears attenuated among boys, implying that being male may buffer the prospective link between low self-esteem and later victimization.

Implication

Theoretically, the current findings both validate and extend the trauma and victimization adaptation model (McCann et al., 1988). First, we broaden the psychological adaptation outcomes of experiencing an adverse/negative life experience to include prosocial behavior as a form of adaptive behavioral response. The study demonstrates that bullying victimization, as an adverse/negative life experience with potentially traumatic features, may initiate a vicious cycle involving prosocial behavior, but this relationship is observed only in boys. The findings suggest that investigations into the links between adverse/negative life

experiences and psychological adaptation should consider both reciprocal effects and potential gender differences.

Furthermore, this study establishes for the first time the potential longitudinal mediating role of self-esteem, conceptualized as a schema related to the self, in the link between bullying victimization and prosocial behavior. Results further emphasize the significance of self-esteem within the trauma and victimization adaptation model (McCann et al., 1988). Through self-esteem, individuals not only sustain impairments to their self-concept during an adverse/negative life experience, but these changes also influence their psychological adaptation, such as a reduction in prosocial behavior. It is also noteworthy that the longitudinal mediation effect was observed only in girls, highlighting the significance of taking gender into account while analyzing self-schema as a mechanism.

Practically, the gender-specific pathways observed in this study point to the value of adopting differentiated strategies: reinforcing prosocial competencies may help boys break the cycle of victimization, whereas bolstering self-esteem could be particularly beneficial for girls in reducing vulnerability to social and psychological risks. Although previous research has emphasized that boys tend to place greater importance on self-esteem (McMullin & Cairney, 2004; Moksnes et al., 2010; Young & Mroczek, 2003), our results reveal that self-esteem is equally, and potentially even more, crucial for girls in the context of bullying victimization and prosocial behavior. Collectively, these findings highlight the necessity of integrating gender-sensitive components into school-based anti-bullying and mental health programs to ensure that interventions effectively address the distinct developmental needs of both boys and girls.

Limitation and Future Direction

While the present study advances knowledge of promoting positive adolescent social development, several limitations warrant further exploration in future research. Firstly, all measures were obtained from adolescents' self-reports, involving response biases. Future research should employ multi-method assessment designs, such as incorporating implicit self-esteem paradigms or experimental tasks for prosocial behavior (Buhrmester et al., 2011; Grumm et al., 2009; Piff et al., 2010).

Secondly, the current study was unable to further differentiate among distinct roles in peer bullying, such as pure victims, individuals with dual roles as both bullies and victims. It may bias the direction and size of the estimated cross-lagged associations, particularly if bully-victims are overrepresented in the sample. In addition, the current study did not identify bystanders in bullying contexts. Increasing research linked defending behavior with bullying

victimization, and this association might be gender specific (i.e., girls tended to engage in supportive and help-seeking forms of defending, such as providing comfort and informing authorities, whereas boys more often adopted aggressively defending; Lambe & Craig, 2020). Future research would benefit from incorporating typological information on bullying roles to more systematically examine the psychological and behavioral responses associated with different roles in bullying situations and further test gender differences in the association.

Thirdly, experiences of victimization differ accordingly. The current study adopted a unidimensional measure of bullying victimization, which precluded examination of potential differences across subtypes, such as overt, relational, and reputational victimization (Wen et al., 2023). Gendered socialization may lead to inconsistent relations between different forms of bullying victimization and psychosocial outcomes among boys and girls (Zhao et al., 2025b), underscoring the importance of further exploring the role of gender in future research.

Fourthly, the number of schools included in the current study was relatively small, which limited the feasibility of treating schools as a cluster for a multilevel RI-CLPM that distinguishes between individual variance and school-level variance. Schools were dummy-coded and entered as control variables in the analyses rather than being modeled as a cluster. Although this practice enabled us to partially account for school-level differences, it may not fully capture the variability across schools. Future studies should include a larger number of schools, which would enable schools to be modeled as clusters and allow for a more rigorous approach to separating school-level effects on bullying victimization and prosocial behavior.

Lastly, all data were collected within a Chinese sociocultural setting. Therefore, the extrapolation of our findings for non-Chinese adolescents remains unclear. Future research is encouraged to incorporate cross-cultural samples in the longitudinal paths. In addition, Chinese culture highly values maintaining harmonious group relations and conformity to collective expectations (Yang et al., 2019). Moreover, Chinese culture is influenced by Confucian ideals, modesty, and societal expectations, which can shape individuals' prosocial actions and their self-perceptions (Cai et al., 2011). Additional studies could examine the role of social norms and gendered expectations among Chinese samples to explore whether the observed gender differences can be explained by culture-specific norms or expectations.

Conclusion

Clarifying the reciprocal association between bullying victimization and prosocial behavior in early adolescence requires disentangling stable between-person differences to more precisely capture time-varying within-person effects. Prior research has typically emphasized unidirectional effects and has rarely distinguished between-person from within-person processes. Furthermore, empirical evidence remains insufficient regarding the mediating role of self-esteem in the reciprocal association between bullying victimization and prosocial behavior, as well as potential gender differences in these processes. This study employed a multigroup RI-CLPM to examine the within-person effects of bullying victimization, self-esteem, and prosocial behavior among Chinese boys and girls, and to test the mediating role of self-esteem. The findings indicated that, at the within-person level, the reciprocal relationship between bullying victimization and prosocial behavior was significant only among boys. In contrast, among girls, the reciprocal association operated only indirectly via self-esteem. More specifically, bullying victimization undermined self-esteem, which was associated with reduced prosocial behavior; conversely, lower prosocial behavior predicted declines in self-esteem, further contributing to increased bullying victimization.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10964-026-02329-6>.

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Authors' Contributions HB conceived the study, participated in its design, performed statistical analysis, coordinated and drafted the manuscript; LL conceived the study, participated in its design, collected data, coordinated and reviewed the manuscript; YB conceived the study, participated in its design, collected data, and reviewed the manuscript; MZ interpreted the data, drafted and reviewed the manuscript; All authors read and approved the final manuscript.

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Declarations

Ethical Approval All procedures performed in the present study were in accordance with the recommendations of the Research Ethics Committee of the Beijing Normal University Collaborative Innovation Center of Assessment toward Basic Education Quality and with the Declaration of Helsinki.

Data Sharing Declaration The datasets generated and/or analyzed dur-

ing the current study are not publicly available but are available from the corresponding author on reasonable request.

Informed Consent Informed consent was obtained from all individual participants included in the study.

Conflict of Interest The authors declare no competing interests.

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